Machine – Calibrate

Menu: Machine > Calibrate

Use this screen to determine the **Cal Factor** (pulses per unit) for a product. The goal is to achieve a consistent meter roller RPM or liquid flow rate matching typical field operation.

Step-by-Step Calibration

① Step 1 — Set Ground Speed

- Enter the ground speed you typically operate at.

② Step 2 — Set Meter Roller RPM / Valve Position

- Switch the product **ON** using the power button.
- Enter an **Initial Base Rate** and an **Estimated Cal Factor**. The Base Rate becomes the product's working rate when Save is pressed.
- Press **Start** — RC adjusts flow to match the target rate.
- If successful, RPM is locked.
- The PWM value used is shown in the **PWM** box (for motor controllers). While locked and stopped it can be edited — accept the value RC found, or enter your own (0–255).
- RC stops automatically when the target rate is met.

If RC fails to reach target RPM:

- Press **Stop**.
- Check ground speed.
- Adjust the estimated Cal Factor, PWM min/max, or metering drive range.

③ Step 3 — Set Actual Cal Factor

- Press **Start** to run the meter roller and collect a sample.
- Press **Stop**, then enter the measured amount collected.
- RC calculates and displays the new Cal Factor.
- Press **Save** to record it.

④ Step 4 — Optional Adjustments

- To redo meter speed: press **Unlock**, then repeat Step 2.
- To reuse a previously saved setting: press **Lock**, then proceed to Step 3.
- To skip Step 2 entirely (motor controllers): press **Lock**, enter the desired **PWM** directly, then proceed to Step 3. Useful for iterating on the metering-unit opening at a fixed PWM — run, weigh, adjust the opening, repeat.
- The lock is saved with Save and remembered the next time the product is calibrated, so Step 3 can be repeated without redoing Step 2.

Calibrating from a Phone

The **Calibration** page on the phone runs the same calibration as this screen, so Start, Stop and the measured amount can be entered standing at the meter or the catch container. To connect a phone, see File > Options > Phone.

- The page has the product buttons 1–5 (lit when that product's power is on), **Power**, the lock (shown as **Meter set** / **Meter not set**), **Start** / **Stop**, the live Pulses and Hz, and the Speed, Base Rate, Meter Cal, Expected, Measured and PWM boxes, with **Save** / **Cancel**.
- Tap a box to enter a value. The old value is shown faded; tapping away without typing keeps it.
- The tablet and the phone show the same calibration, and either can start or stop it. This screen follows a run started from the phone.
- **A run started from the phone stops by itself** if the phone loses its connection, or its screen is locked or switched to another app, for about 5 seconds. The page then shows *Last run stopped: phone connection lost*.
- **A run started on the tablet does not stop by itself.** Start from the phone if you will be away from the tablet. Stop works from either.
- Closing this screen stops a run started on the tablet; a run started from the phone keeps going.

Controls

Control	Action
Start	Start the meter roller / begin collecting a sample
Stop	Stop the meter roller
Lock/Unlock	Lock or unlock the current RPM setting. Shown on the phone as Meter set / Meter not set. Saved with Save
Power on/off	Switch the product on or off
Cancel	Discard changes
Save	Save the calculated Cal Factor
PWM	PWM used while locked (motor controllers only). Editable when locked and stopped; press Save to keep the value
Hz	Live sensor pulse frequency for the selected product (shown for all control types)

Important Notes

- AgOpenGPS is **not** used during calibration.
- The Switchbox is **not** used during calibration.
- If a switchbox is present, ensure the **master switch is OFF** before calibrating.
- Calibration uses **Constant UPM, Mode 2**.
- All sections on the calibrating module are activated during calibration; modules that are not calibrating are unaffected.
- More than one product can be calibrated at the same time, on the same or different modules (requires up-to-date module firmware).
- After calibration, the original mode and section states are restored.

- If RateController closes during a calibration (power loss, a crash, or ended in Task Manager), the product's original settings and the speed source are restored at the next start, with the message *Product N restored after an interrupted calibration*. A calibration that wasn't saved has to be repeated.
- Start is refused if a locked motor product has PWM 0 — enter a PWM value first, or the motor would not turn.
- Watch the **Hz** display while running: aim for at least 10–12 Hz at working rate for a stable reading. Very high frequencies can exceed the sensor's switching limit — if Hz plateaus while speed rises, the sensor is at its ceiling.